

Itai Lang

Computer Graphics • Computer Vision • Machine Learning

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Academic Appointment

University of Chicago

Chicago, Illinois

POSTDOCTORAL RESEARCH FELLOW

July 2022 - Present

- Host: Prof. Rana Hanocka

Education

PhD Electrical Engineering, Tel Aviv University, Israel

November 2017 - June 2023

- ADVISOR: PROF. SHAI AVIDAN
- Topic: New Frontiers in Learning-Based Processing of 3D Point Clouds

MSc Electrical Engineering, Tel Aviv University, Israel

November 2006 - June 2013

- ADVISORS: DR. HEDVA SPITZER AND PROF. NAHUM KIRYATI
- Topic: Lesion Segmentation in Medical Images

BSc Electrical Engineering, The Technion, Israel

November 2001 - June 2005

BA Physics, The Technion, Israel

- SUMMA CUM LAUDE
- Topic: Statistical Analysis of Medical Images

Publications

JOURNAL ARTICLES & CONFERENCE PROCEEDINGS

- [1] **Best Segmentation Buddies for Image-Shape Correspondence**
Itai Lang*, Dongwei Lyu*, Dale Decatur, Rana Hanocka
Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2026 [CVF]
- [2] **Deep Feature Deformation Weights**
Richard Liu, Itai Lang, Rana Hanocka
Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2026 (Award Candidate) [CVF]
- [3] **Structurally Disentangled Feature Fields Distillation for 3D Understanding and Editing**
Yoel Levy, David Shavin, Itai Lang, Sagie Benaim
Proceedings of the International Conference on 3D Vision (3DV), 2026 [OpenReview]
- [4] **WIR3D: Visually-Informed and Geometry-Aware 3D Shape Abstraction**
Richard Liu, Daniel Fu, Noah Tan, Itai Lang, Rana Hanocka
Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025 (Oral) [CVF]
- [5] **Geometry in Style: 3D Stylization via Surface Normal Deformation**
Nam Anh Dinh, Itai Lang, Hyunwoo Kim, Oded Stein, Rana Hanocka
Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2025 [CVF]
- [6] **GeoCode: Interpretable Shape Programs**
Ofek Pearl, Itai Lang, Yuhua Hu, Raymond A. Yeh, Rana Hanocka
Computer Graphics Forum, Volume 44, Number 1, 2025 [Wiley]
- [7] **iSeg: Interactive 3D Segmentation via Interactive Attention**
Itai Lang, Fei Xu, Dale Decatur, Sudarshan Babu, Rana Hanocka
SIGGRAPH Asia 2024 Conference Papers, 2024 [ACM]

- [8] **MeshUp: Multi-Target Mesh Deformation via Blended Score Distillation**
Hyunwoo Kim, **Itai Lang**, Noam Aigerman, Thibault Groueix, Vladimir G Kim, Rana Hanocka
Proceedings of the International Conference on 3D Vision (3DV), 2025 [[arXiv](#)]
- [9] **DGD: Dynamic 3D Gaussians Distillation**
Isaac Labe, Noam Issachar, **Itai Lang**, Sagie Benaim
Proceedings of the European Conference on Computer Vision (ECCV), 2024 [[Springer](#)] [[arXiv](#)]
- [10] **HaLo-NeRF: Learning Geometry-Guided Semantics for Exploring Unconstrained Photo Collections**
Chen Dudai, Morris Alper, Hana Bezalel, Rana Hanocka, **Itai Lang**, Hadar Averbuch-Elor
Computer Graphics Forum, Volume 43, Number 2, 2024 [[Wiley](#)]
- [11] **3D Paintbrush: Local Stylization of 3D Shapes with Cascaded Score Distillation**
Dale Decatur, **Itai Lang**, Kfir Aberman, Rana Hanocka
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024 [[CVF](#)]
- [12] **3D Highlighter: Localizing Regions on 3D Shapes via Text Descriptions**
Dale Decatur, **Itai Lang**, Rana Hanocka
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023 (Highlight, top 2.5%) [[CVF](#)]
- [13] **SCOOP: Self-Supervised Correspondence and Optimization-Based Scene Flow**
Itai Lang, Dror Aiger, Forrester Cole, Shai Avidan, Michael Rubinstein
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023 [[CVF](#)]
- [14] **SAGA: Spectral Adversarial Geometric Attack on 3D Meshes**
Tomer Stolik*, **Itai Lang***, Shai Avidan
Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023 [[CVF](#)]
- [15] **Co-occurrence Based Texture Synthesis**
Anna Darzi, **Itai Lang**, Ashutosh Taklikar, Hadar Averbuch-Elor, Shai Avidan
Computational Visual Media, Volume 8, Number 2, 2022 [[IEEE](#)]
- [16] **DPC: Unsupervised Deep Point Correspondence via Cross and Self Construction**
Itai Lang*, Dvir Ginzburg*, Shai Avidan, Dan Raviv
Proceedings of the International Conference on 3D Vision (3DV), 2021 [[IEEE](#)] [[arXiv](#)]
- [17] **Geometric Adversarial Attacks and Defenses on 3D Point Clouds**
Itai Lang, Uriel Kotlicki, Shai Avidan
Proceedings of the International Conference on 3D Vision (3DV), 2021 [[IEEE](#)] [[arXiv](#)]
- [18] **SampleNet: Differentiable Point Cloud Sampling**
Itai Lang, Asaf Manor, Shai Avidan
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020 (Oral, top 5.2%) [[CVF](#)]
- [19] **Learning to Sample**
Oren Dovrat*, **Itai Lang***, Shai Avidan
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019 [[CVF](#)]
- [20] **Multi-scale Texture-based Level-set Segmentation of Breast B-mode Images**
Itai Lang*, Miri Sklair-Levy*, Hedva Spitzer
Computers in Biology and Medicine, Volume 72, 2016 [[Elsevier](#)]

WORKS UNDER SUBMISSION

- [21] **LL3M: Large Language 3D Modelers**
Sining Lu, Guan Chen, Nam Anh Dinh, **Itai Lang**, Ari Holtzman, Rana Hanocka
arXiv preprint arXiv:2508.08228, 2025 [[arXiv](#)]
- [22] **3D PixBrush: Image-Guided Local Texture Synthesis**
Dale Decatur, **Itai Lang**, Kfir Aberman, Rana Hanocka
arXiv preprint arXiv:2507.03731, 2025 [[arXiv](#)]

PATENT

- **Event-Based Sensor that Filters for Flicker**
Itai Lang, Nathan Henri Levy
US Patent Application 15/988,005, 2019 [[US 2019/0362256 A1](#)]

Experience

University of Chicago

POSTDOCTORAL RESEARCH FELLOW

Chicago, Illinois

July 2022 - Present

- **Conducting independent research in 3D computer graphics**, focusing on geometry analysis and generative modeling
- **Publishing multiple papers** in top-tier graphics and vision venues and **delivering invited talks** internationally
- **Mentoring graduate and undergraduate students** with no prior research experience, resulting in publications in prestigious venues
- **Contributing to competitive grant proposals** with host faculty, providing feedback on research plans, and preparing supporting materials
- **Served as the group leader** for six months during faculty leave (2024), mentoring students and managing lab activities
- **Collaborating across academic and industrial research institutions** (Cornell University, Columbia University, Adobe Research) on diverse projects in 3D graphics and vision
- **Initiated independent collaboration** outside the host lab with Prof. Sagie Benaim, researching 3D scene understanding and manipulation
- **Organized the first Great Lakes Graphics Workshop (2025)**, bringing together researchers from multiple universities in the region
- **Authored a widely used guide** for creating academic conference posters, covering design, content, and presentation best practices
- **Created the lab's onboarding guide**, detailing software setup, server access, example project workflows, and debugging practices; adopted by all new members

Tel Aviv University

DOCTORAL RESEARCH FELLOW

Tel Aviv, Israel

November 2017 - June 2023

- **Pioneered task-oriented learned point cloud sampling**, establishing a widely cited research direction that inspired numerous follow-up studies
- **Developed efficient methods for 3D point cloud correspondence**, advancing the state-of-the-art in non-rigid shape matching and scene flow estimation
- **Released open-source, reproducible code** alongside publications, accelerating adoption in the research community
- **Mentored Master's students** from their first research experience through graduation, resulting in joint publications in top-tier venues
- **Invited to present the research** at multiple universities, seminars, and companies

Google Research

RESEARCH INTERN

Tel Aviv, Israel

August 2021 - June 2022

- **Researched self-supervised scene flow estimation** based on point correspondence and residual refinement optimization
- The research resulted in a **publication at CVPR**, the top-tier computer vision conference

Samsung Israeli Research Center

RESEARCH ENGINEER

Tel Aviv, Israel

June 2013 - August 2021

- **Developed hand detection and gesture recognition methods** for the Dynamic Vision Sensor
- **Invented an anti-flicker algorithm** for the Dynamic Vision Sensor, based on a Hidden Markov Model; resulted in a patent publication
- Developed methodology and infrastructure for research procedures, algorithm development standards, and automatic code integration
- Train junior engineers and act as a center of knowledge

Tel Aviv University

GRADUATE RESEARCH FELLOW

Tel Aviv, Israel

November 2006 - June 2013

- **Researched level-set segmentation of ultrasound breast images**
- Collaborated with the head of the Breast Imaging Unit at Sheba Medical Center

Matzov, The IDF Data Security Center

RESEARCH OFFICER

Tzrifin Base, Israel

November 2006 - June 2012

- **Researched Image forgery detection** using the Expectation-Maximization technique
- Developed information filtering systems based on statistical discrimination learning
- Researched information leakage mechanisms and designed prevention solutions

Honors & Awards

Best Paper Honorable Mention at the International Conference on 3D Vision (3DV)	2025
Accepted to the Doctoral Consortium at CVPR	2023
Holon City Mayor's Award for Academic Excellence	2005
The Technion President's Academic Excellence Award	2005
The Technion President's Academic Excellence Award	2004
The Technion Scholarship for Outstanding Candidates	2001
Accepted to PSAGOT, the elite academic program of the Israel Defense Forces	2001

Invited Talks

Clicks and Prompts: Intuitive Interactive Segmentation and Local Editing of 3D Shapes

2025	University of Toronto — Host: Prof. David B. Lindel (Virtual)	Toronto, Canada
2025	Visual Computing Seminar , Brown University — Host: Prof. Daniel Ritchie	Providence, Rhode Island
2025	Simon Fraser University — Host: Prof. Hao Richard Zhang (Virtual)	Vancouver, Canada

iSeg: Interactive 3D Segmentation via Interactive Attention

2025	Great Lakes Graphics Workshop	Chicago, Illinois
2024	Computer Vision Forum, Tel Aviv University	Tel Aviv, Israel
2023	Graphics and Vision Seminar, Toyota Technological Institute at Chicago (TTIC)	Chicago, Illinois

Flexible and Intuitive Region Localization on 3D Shapes

2023	OriginAI — Host: Rami Ben-Ari	Tel Aviv, Israel
2023	Vision Seminar, The Hebrew University of Jerusalem — Host: Prof. Shmuel Peleg	Jerusalem, Israel

3D Highlighter: Localizing Regions on 3D Shapes via Text Descriptions

2023	Computer Vision Forum, Tel Aviv University	Tel Aviv, Israel
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New Frontiers in Learning-Based Processing of 3D Point Clouds

2023	Electrical Engineering Department Seminar, Tel Aviv University	Tel Aviv, Israel
2022	Computer Science Department Seminar, University of Chicago	Chicago, Illinois

Pure Point Correspondence for Decoderless Shape Matching and Hybrid 3D Scene Flow

2022	Israel Computer Vision Day — Hosts: Prof. Shai Avidan & Prof. Raja Giryes	Tel Aviv, Israel
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SCOOP: Self-Supervised Correspondence and Optimization-Based Scene Flow

2022	Computer Vision Forum, Tel Aviv University	Tel Aviv, Israel
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DPC: Unsupervised Deep Point Correspondence via Cross and Self Construction

2022	Advanced Graphics Seminar, Tel Aviv University — Host: Prof. Daniel Cohen-Or	Tel Aviv, Israel
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Geometric Adversarial Attacks and Defenses on 3D Point Clouds

2022	Machine Learning Seminar, Bar Ilan University	Ramat Gan, Israel
2021	Israel Computer Vision Day — Hosts: Prof. Shai Avidan & Prof. Raja Giryes	Tel Aviv, Israel
2021	Pixel Club Seminar, The Technion	Haifa, Israel
2021	Israeli Geometric Deep Learning Workshop (Virtual)	Tel Aviv, Israel

Learned Sampling of 3D Point Clouds

2021	Pixel Club Seminar, The Technion (Virtual)	Haifa, Israel
2020	Visual Computing Seminar, Tel Aviv University — Host: Prof. Daniel Cohen-Or	Tel Aviv, Israel
2020	Alibaba Academy Meetup (Virtual)	Tel Aviv, Israel
2020	General Motors	Tel Aviv, Israel
2019	Israel Computer Vision Day — Hosts: Prof. Shai Avidan & Prof. Raja Giryes	Tel Aviv, Israel

SampleNet: Differentiable Point Cloud Sampling

2020	Israeli Geometric Deep Learning Workshop (Virtual)	Tel Aviv, Israel
2020	CVPRi Workshop — Host: The Machine & Deep learning Israel Group (Virtual)	Tel Aviv, Israel

Learning to Sample

2019	Vision Seminar, The Hebrew University of Jerusalem — Host: Prof. Shmuel Peleg	Jerusalem, Israel
2019	AI and Cyber Day, Tel Aviv University	Tel Aviv, Israel
2019	Samsung Israeli Research Center	Ramat Gan, Israel
2019	Deep Learning Seminar, Ben-Gurion University — Host: Prof. Tammy R. Raviv	Be'er Sheva, Israel

Mentoring

Research Supervisor & Academic Mentor, University of Chicago

PHD THESIS

- Richard Liu - shape abstraction, shape deformation, 2024 - present
- Nam Anh Dinh - geometric stylization, generative deformation and remeshing, 2023 - present
- Dale Decatur - neural localization and texturing, 2022 - present

MASTER THESIS

- Sining Lu - LLMs for 3D asset generation, 2025 (ongoing)
- Raj Hansini - dynamic texture synthesis, 2025 (ongoing)

BACHELOR RESEARCH

- Albert Seo - 3D textur transfer, 2024 - present
- Hyunwoo Kim - multi-target shape deformation, asset-body alignment, 2023 - present

RESEARCH INTERNSHIP

- Harper Lyu - image-shape correspondence, 2025 (ongoing)
- Fei Xu - interactive 3D segmentation, 2023

Research Supervisor & Academic Mentor, Tel Aviv University

MASTER THESIS

- Ofek Pearl - interpretable shape editing, 2023 - 2024
- Chen Dudai - 3D scene segmentation, 2023
- Tomer Stolik - spectral geometric attacks, 2022
- Anna Darzi - texture synthesis, 2019
- Asaf Manor - differentiable point cloud sampling, 2019
- Oren Dovrat - learning point cloud sampling, 2018

MASTER PROJECT

- Uriel Kotlicki - geometric attacks on point clouds, 2020

BACHELOR PROJECT

- Royi Avron & Amit Shomer - local point cloud sampling, 2021

Teaching

[CMSC 3540] Topics in Machine Learning: 3D Geometry Processing and Computer Vision with Deep Neural Networks

University of Chicago

TEACHING ASSISTANT

Winter 2024

- Number enrolled: 10

[0510-6251-01] Computer Vision

Tel Aviv University

LEAD LECTURER

Fall 2024

- Number enrolled: 115

Service

Professional Activities

- Workshop Lead Organizer, [Great Lakes Graphics 2025](#)
- International Program Committee, SGP 2025
- Area Chair, WACV 2025
- Student Research Competition Judge, SIGGRAPH 2024
- Poster Juror, SIGGRAPH 2024

Reviewer

- CVPR 2020, 2023 (Outstanding Reviewer), 2024, 2025 (Outstanding Reviewer), 2026
- SIGGRAPH Asia 2023, 2024, 2025, 2026
- SIGGRAPH 2022, 2024, 2025, 2026
- ECCV 2026
- ICLR 2025
- 3DV 2025
- Graphics Replicability Stamp Initiative 2024
- Transactions on Visualization and Computer Graphics TVCG 2021
- Transactions on Circuits and Systems for Video Technology TCSVT 2021
- Computers & Graphics CAG 2021
- Transactions on Pattern Analysis and Machine Intelligence TPAMI 2021

Media Coverage

[Featured article](#) on the Great Lakes Graphics Workshop, *University of Chicago CS News*

April 2025

Presented the work *SampleNet* at the [Talking Papers Podcast](#)

March 2022

Ongoing Collaborations

Prof. Ari Holtzman

UNIVERSITY OF CHICAGO

- Innovative 3D asset generation via Large Language Model Agents

Prof. Oded Stein

UNIVERSITY OF SOUTHERN CALIFORNIA

- Deep neural geometric stylization, generative deformation and remeshing

Prof. Hadar Averbuch-Elor

CORNELL UNIVERSITY

- 3D asset-body alignment

Prof. Silvia Sellán

COLUMBIA UNIVERSITY

- 3D asset-body alignment

Prof. Sagie Benaim

THE HEBREW UNIVERSITY OF JERUSALEM

- Learning 4D scene understanding and editing

Skills

Languages: Hebrew (native speaker), English (proficient)

Programming languages: Python, PyTorch, Tensorflow, C/C++

Hobbies: Capoeira (Israel Champion, 2022), tropical fruit gardening, running